

NEUROMUSCULAR JUNCTION

Headings

- ▼ NEUROMUSCULAR JUNCTION
 - AXON TERMINAL AND MOTOR ENDPLATE
 - SYNAPTIC TROUGH OR GUTTER
 - SYNAPTIC CLEFT
 - SUBNEURAL CLEFTS
- ▼ NEUROMUSCULAR TRANSMISSION
 - Events of Neuromuscular Transmission:
 - SEQUENCE OF EVENTS (FLOWCHART)
- ▼ DETAILED EVENTS
 1. Release of ACh
 2. Action of ACh
 3. Development of Endplate Potential
 4. Development of Miniature Endplate Potential (MEPP)
 5. Destruction of Acetylcholine
- NEUROMUSCULAR BLOCKERS

- **Definition** : Junction between terminal branch of nerve fiber and muscle fiber.
- **Structure** : Skeletal muscle fiber innervated by motor nerve fiber.

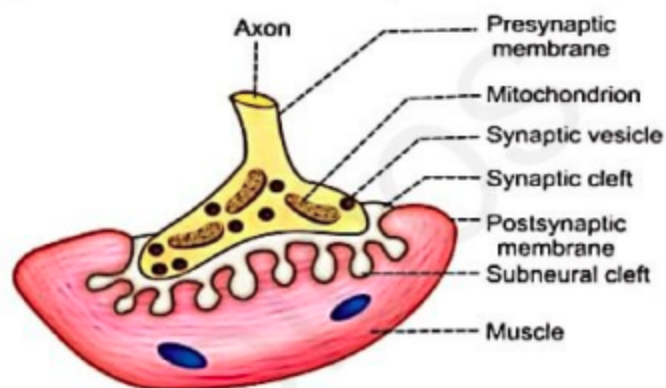


FIGURE 32.2: Structure of neuromuscular junction

AXON TERMINAL AND MOTOR ENDPLATE

- Terminal branch of nerve fiber.

- **When axon comes close to muscle fiber :**
 - It loses myelin sheath.
 - Axis cylinder exposed (*this portion expanded like a bulb*) → **called motor endplate.**
- **Axon Terminal Contains :—**
 - a. **Mitochondria** : Contain ATP (*source of energy for synthesis of ACh*).
 - b. **Synaptic Vesicles** : Contain neurotransmitters, acetylcholine (ACh) synthesized.

SYNAPTIC TROUGH OR GUTTER

- Motor endplate invaginates inside muscle fiber → **forms a depression (gutter).**

SYNAPTIC CLEFT

- Membrane of nerve ending → **Presynaptic membrane.**
- Membrane of muscle fiber → **Postsynaptic membrane.**
- **Synaptic Cleft** : Space between these 2 membranes.
 - Contains **basal lamina** (*thin layer of spongy reticular matrix through which extracellular fluid diffuses*).
 - **Acetylcholinesterase (AChE)** is attached to matrix of basal lamina.

SUBNEURAL CLEFTS

- Postsynaptic membrane thrown into **numerous folds.**
- Postsynaptic membrane contains receptors called **nicotinic acetylcholine receptors.**

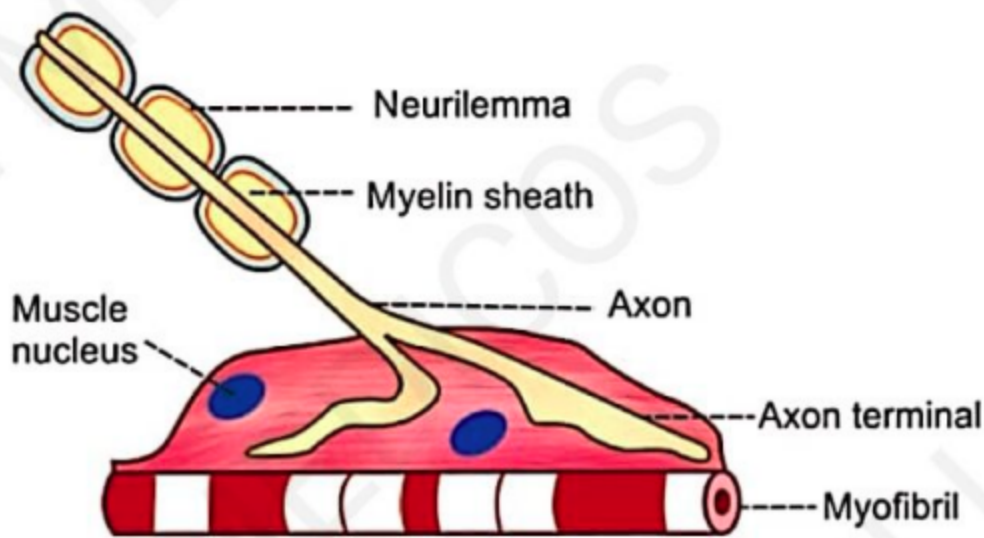


FIGURE 32.1: Longitudinal section of neuromuscular junction

NEUROMUSCULAR TRANSMISSION

- **Definition :** The transfer of information from motor nerve ending to muscle fiber through neuromuscular junction.
 - Mechanism by which motor nerve impulses initiate muscle contraction.

Events of Neuromuscular Transmission:

- a. Release of acetylcholine
- b. Action of acetylcholine
- c. Development of endplate potential
- d. Development of miniature endplate potential
- e. Destruction of acetylcholine

SEQUENCE OF EVENTS (FLOWCHART)

Motor Nerve Fiber & Axon Terminal

Action potential in motor
nerve fiber



Opening of voltage-gated
calcium channels



Entry of calcium ions
(Ca^{2+}) from ECF into axon
terminal



Opening / bursting of
vesicles & release of ACh
by exocytosis



Synaptic Cleft

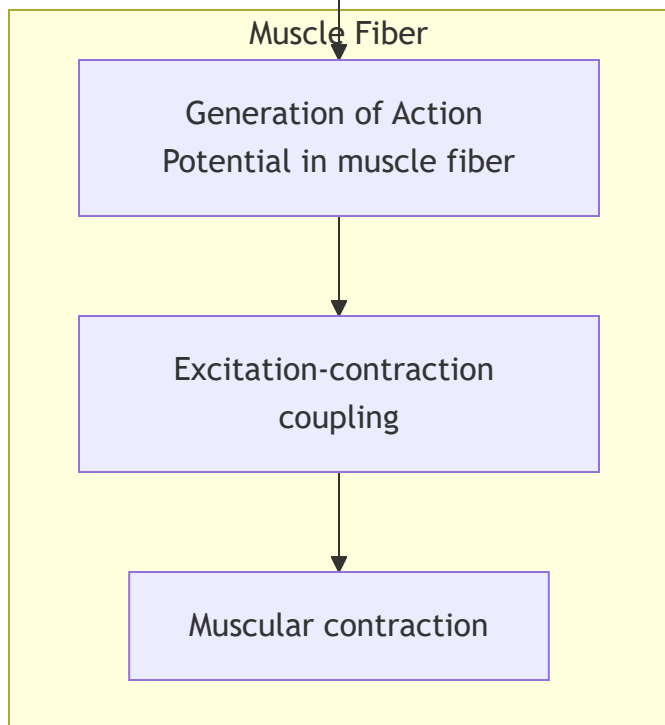
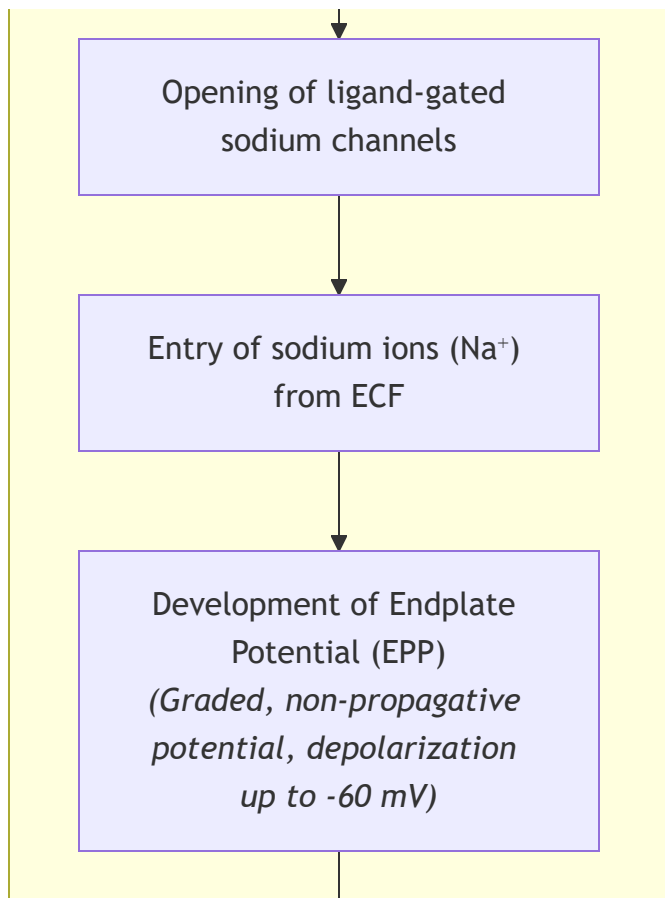
Passage of ACh across
synaptic cleft



Postsynaptic Membrane

Binding of ACh with
nicotinic receptors
(Formation of ACh-
Receptor complex)





DETAILED EVENTS

1. Release of ACh

- Ca^{2+} ions cause bursting of vesicles by forcing synaptic vesicles to move and fuse with presynaptic membrane.
- By **exocytosis**, acetylcholine diffuses into synaptic cleft.

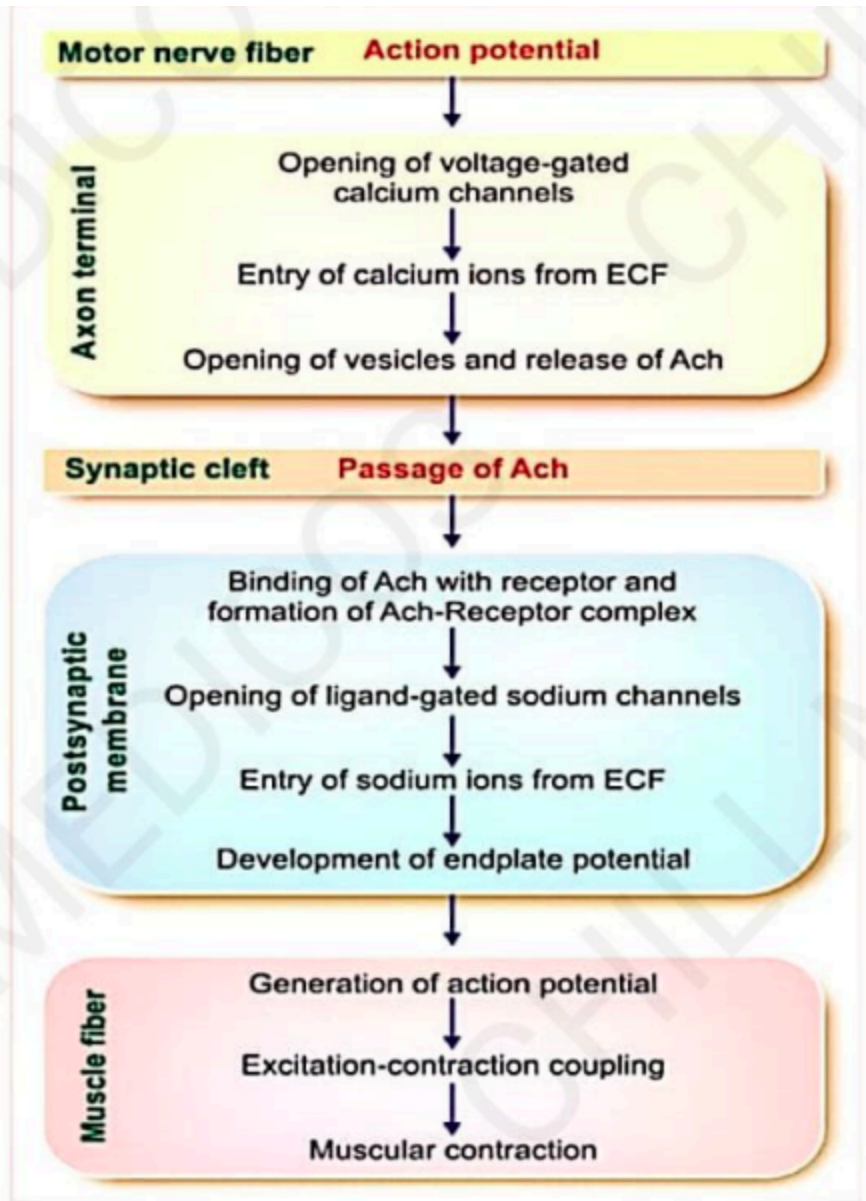


FIGURE 32.3: Sequence of events during neuromuscular transmission. ACh = Acetylcholine, ECF = Extracellular fluid.

2. Action of ACh

- After entering cleft, ACh molecules bind with **nicotinic receptors** present in postsynaptic membrane (*ACh–Receptor complex*).
- Increases permeability of postsynaptic membrane for Na^+ by **opening ligand-gated Na^+ channels**.

3. Development of Endplate Potential

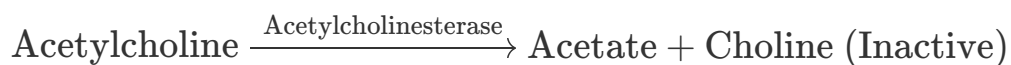
- The change in RMP when impulse reaches neuromuscular junction.
- Slight depolarisation up to -60 mV (*as Na^+ ions enter*).
- **Endplate Potential (EPP)** : Graded potential, non-propagative (*causes development of action potential in muscle fiber*).

4. Development of Miniature Endplate Potential (MEPP)

- A weak endplate potential in neuromuscular junction that is developed by the **release of ACh from axon terminal**.
- **Amplitude** : 0.5 mV .
- Miniature endplate potential cannot produce action potential in muscle alone.
- When ACh is released continuously $\longrightarrow$ miniature endplate potentials are **added together** and finally produce endplate potential $\longrightarrow$ **results in action potential**.

5. Destruction of Acetylcholine

- Acetylcholine released is **destroyed very quickly** (within 1 msec) by **acetylcholinesterase (AChE)**.
- Prevents repeated excitation of muscle fiber $\longrightarrow$ **allows muscle to relax**.



Taken back into axon terminal by reuptake process

NEUROMUSCULAR BLOCKERS

- **Definition :** Drugs which prevent transmission of impulses from nerve fibers to muscle fiber through neuromuscular junctions.
- **Examples & Mechanisms :—**
 - a. **Curare :** By combining with ACh receptors.
 - b. **Bungarotoxin :** By blocking ACh receptors.
 - c. **Botulinum Toxin :** Prevents release of ACh.
 - d. **Succinylcholine & Carbamylcholine :** Act like excess ACh (*depolarizing blockers*).